Exercise 4.1

1. (i) $x = n\pi,\ n = 0, \pm1, \pm2, \ldots \pm10$ (ii) $x = (4n-1)\frac{\pi}{2},\ n = 0, \pm1$

2. (i) 1, $\frac{2\pi}{7}$ (ii) 1, 6π (iii) 4, π 4. (i) $\frac{\pi}{3}$ (ii) $-\frac{\pi}{4}$

5. $x = 0$ 6. (i) $\{-1, 1\}$ (ii) $[0, 1]$ 7. $\frac{\pi}{3}$

Exercise 4.2

1. (i) $x = (2n+1)\frac{\pi}{2},\ n = 0, \pm1, \pm2, \pm3, \pm4, \pm5, -6$ (ii) $x = (2n+1)\pi,\ n = 0, \pm1, \pm2, -3$

2. $-\frac{\pi}{6} \notin [0, \pi]$ 3. True 4. $\frac{\pi}{3}$

5. (i) $\frac{5\pi}{6}$ (ii) $-\frac{\pi}{6}$ (iii) $\frac{24\pi}{119}$ 6. (i) $[-5, 5]$ (ii) $[-1, 1]$

7. $0 < x < \frac{1}{3}$ 8. (i) 0 (ii) $\frac{17\pi}{12}$

Exercise 4.3

1. (i) $[-3, 3]$ (ii) $\mathbb{R}$ 2. (i) $\frac{\pi}{4}$ (ii) $-\frac{\pi}{6}$

3. (i) $\frac{7\pi}{4}$ (ii) 1947 (iii) –0.2021 4. (i) ∞ (ii) $-\frac{2\sqrt{5}}{25}$ (iii) $\frac{24}{25}$

Exercise 4.4

1. (i) $\frac{\pi}{6}$ (ii) $\frac{\pi}{6}$ (iii) $-\frac{\pi}{4}$ 2.(i) $-\frac{\pi}{3}$ (ii) $\cot^{-1}(2) - \frac{\pi}{6}$ (iii) $-\frac{5\pi}{6}$

Exercise 4.5

1. (i) $-\frac{\pi}{2}$ (ii) $-\frac{\pi}{4}$ (iii) $5 - 2\pi$

2. (i) $\sqrt{2x - x^2}$ (ii) $\frac{1}{\sqrt{9x^2 - 6x + 2}}$ (iii) $\frac{2x+1}{\sqrt{3 - 4x - 4x^2}}$

3. (i) $\frac{\pi}{6}$ (ii) 0 (iii) $\frac{17}{6}$ 8. $\frac{\pi}{4}$

9. (i) $x = 13$ (ii) $x = \frac{a-b}{1+ab}$ (iii) $x = n\pi + \frac{\pi}{4},\ n \in \mathbb{Z}$

(iv) $x = \sqrt{3}$

10. 3

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